

# Blood Flow Under Wrist Cuff, in Hand Alters Oscillometric Waveforms During Blood Pressure Measurement

Jiri Jilek, MSEE; Teiichiro Fukushima, MD

The oscillometric method evaluates oscillometric waveform amplitudes and cuff pressures to determine systolic, diastolic, and mean arterial pressures. This study investigated the effect of blood flow under the wrist cuff and in the hand on oscillometric waveforms during gradual cuff deflation. The oscillometric waveforms were acquired by a data acquisition system and the reference systolic and diastolic values were obtained by auscultation. The central hypothesis was that the slope (S1) of the oscillometric waveform amplitude envelope at cuff pressures higher than reference systolic pressure and the slope (S3) of the oscillometric waveform amplitude envelope between mean pressure and reference diastolic pressure are steeper than the slope (S2) of the oscillometric waveform amplitude envelope between reference systolic pressure and mean pressure. The decrease in S2 is caused by blood flow under the cuff and in the hand. Thirty-two human subjects were tested while in a sitting position. Results:  $S1 = -0.065 \pm 0.022$ ,  $S2 = -0.025 \pm 0.012$ , and  $S3 = 0.114 \pm 0.035$ . The slopes S1 and S3 were steeper than the slope S2 in all subjects. The results support the central hypothesis. Changes of the slopes have the potential to alter the accuracy of oscillometric blood pressure measurement. Observations of oscillometric waveform contours

revealed that the contours were distorted at cuff pressures higher than reference diastolic pressure but they were not distorted at and below reference diastolic pressure. Algorithms recognizing changes in the slopes and algorithms recognizing the transition from distorted to distortion-free waveform contours could improve accuracy of oscillometric blood pressure determination.

## Introduction

The oscillometric method for noninvasive determination of blood pressure is used in most automatic blood pressure monitors. The method is based on the principle of vascular unloading. Marey in 1876<sup>1</sup> placed the forearm and hand in a water-filled chamber to which a variable counter pressure was applied. The counter pressure for maximum amplitude of oscillations detected in the chamber determined that the vessel walls were maximally relieved of tension at that counter pressure. Posey and Geddes<sup>2</sup> showed that the point of maximal oscillations in an air-inflated cuff corresponded with mean arterial pressure (MAP). When pressure in the cuff was increased above or decreased below MAP, the amplitudes of oscillations decreased. The air-cuff method is similar, but not identical, to Marey's method. While Marey exposed both the forearm and the hand to counter pressure in the chamber, the air-cuff

method exposes only a segment of the limb under the cuff to counter pressure. Oscillations in the cuff are converted into oscillometric waveforms by a pressure sensor, an electronic amplifier, and a filter. Oscillometric waveforms and cuff pressures acquired during a gradual cuff deflation procedure are shown in Figure 1. The waveform amplitudes decrease at cuff pressures above and below MAP and are present at all cuff pressures.

The determination of oscillometric systolic and diastolic pressure values has been controversial. Geddes et al.<sup>3</sup> observed that systolic pressure values were frequently near the cuff pressure value at oscillometric waveform amplitude of about 50% of maximal amplitude and diastolic values at 80% of maximal amplitude. This method of systolic and diastolic pressure determination is called "characteristic ratio method." The values based on the characteristic ratio method were, however, subject to a large intersubject variability. Other methods for systolic and diastolic pressure determination have been introduced. A report describing the function of an oscillometric monitor claims that the device determines systolic pressure as the point of initial increase of the oscillometric waveform amplitudes.<sup>4</sup> Another report puts the systolic pressure value on the minimal ascending slope of the oscillometric waveform amplitude envelope and diastolic pres-

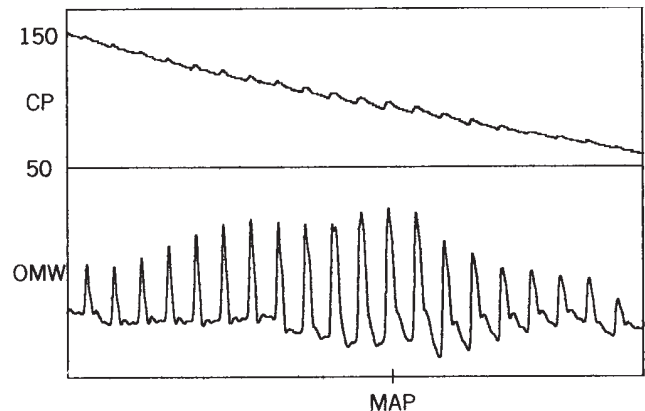
sure on the maximal descending slope of the oscillometric waveform amplitude envelope.<sup>5</sup> The published algorithms for systolic and diastolic pressure determination appear to have been developed empirically<sup>6</sup> and studies linking these algorithms to the underlying circulatory physiology have been lacking. Several investigators published studies of biomechanical models of oscillometric methodology.<sup>7-9</sup> Changes in the oscillometric waveform amplitude envelope slopes under different conditions were observed. The studies concluded that the characteristic ratio method is susceptible to errors. A report by a group of leading experts stressed the importance of accurate blood pressure measurements and called for additional research into the accuracy of oscillometric blood pressure monitors.<sup>10</sup>

The objective of this study was two-fold: (1) to introduce a general hypothesis stating that blood flow under the cuff and in the hand alters oscillometric waveforms during gradual cuff deflation, and (2) to formulate and test a specific hypothesis. The hypothesis states that the slope of the oscillometric waveform amplitude envelope between the reference systolic pressure and mean pressure is less steep than either the slope at cuff pressures higher than reference systolic pressure or the slope between mean pressure and reference diastolic pressure.

## Methods

### Data Acquisition and Reference Blood Pressure Measurements

A specialized data acquisition system developed by the authors was used to acquire cuff pressures, oscillometric waveforms, and finger photoplethysmographic (PPG) data during gradual cuff deflation procedure.<sup>11</sup> The system consists of a wrist cuff; a module containing pneumatic, pressure, and electronic circuits; and a notebook computer with special software. Cuff pressure is converted into analog voltage by a pressure sensor. The analog voltage was amplified by instrumentation and operational amplifiers. The amplified cuff pressure signal was then separated into two channels that are digitized by a 12-bit A/D converter. Sampling rate is 11.6 mS. The gradual cuff deflation was controlled by a current-controlled valve. A microcontroller directly controlled module functions and communicated with the computer via RS-232C interface. A commercial wrist cuff was attached to the left wrist. A small volume of air in the wrist cuff diminished high-frequency damping and facilitated detailed observation of waveform contours. A prototype finger transducer attached to the left index finger was used to acquire PPG

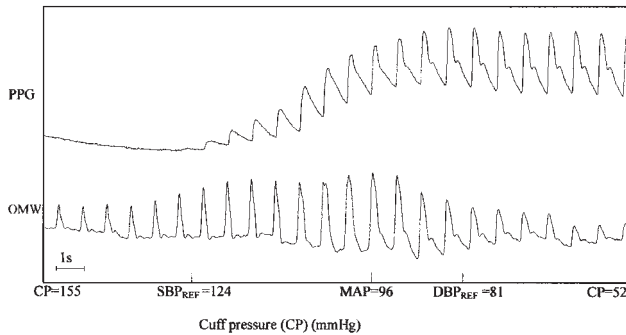


**Figure 1.** Cuff pressures (CP) and oscillometric waveforms (OMW) acquired during a gradual cuff deflation procedure. Mean arterial pressure (MAP) was determined as the point of maximal OMW amplitude.

data for observational purposes. One data acquisition procedure lasted 30 to 40 seconds and was fully automatic. Reference systolic and diastolic pressure values were obtained from the left arm by auscultation with a mercury sphygmomanometer. Auscultatory, sequential blood pressure measurement has been the prevalent method used in validation studies of oscillometric blood pressure monitors. Data collection started with one reference pressure measurement followed after 30 seconds by one automatic data acquisition procedure. Sequential measurements were used in order to not interfere with oscillometric waveform acquisition. Data acquisition procedures and reference blood pressure measurements were performed on 32 subjects while in a sitting position. The oscillometric waveform amplitude and mean pressure values were obtained by software algorithms designed by the authors. The mean pressure values were computed as cuff pressure values for the largest waveform amplitudes.

### Observations of Oscillometric and PPG Waveforms During Gradual Cuff Deflation

A gradual cuff deflation procedure was divided into four segments (A–D). The following section describes oscillometric waveform amplitude and shape changes in each segment in terms of vascular unloading and blood flow. The phases of Korotkoff sounds are mentioned where appropriate. The reference systolic pressure value indicates the approximate beginning of turbulent blood flow under the cuff and the reference diastolic value indicates the approximate beginning of laminar blood flow under the cuff.



**Figure 2.** Oscillometric waveforms (OMWs; bottom trace) and photoplethysmographic (PPG) waveforms (upper trace). Systolic ( $SBP_{REF}$ ) and diastolic ( $DBP_{REF}$ ) are blood pressure reference values obtained by auscultation. MAP is mean arterial pressure determined by a software algorithm. CP is cuff pressure.

**Segment A**—Gradual cuff deflation from cuff pressure values of about 30 mmHg above the reference systolic pressure to the reference systolic pressure (Figure 2).

Oscillometric waveforms are present because the radial artery pulsations are transmitted to the proximal side of the cuff's bladder. The waveform amplitudes increase as cuff pressure is gradually lowered. No blood flow passes under the cuff, the photoplethysmogram baseline is flat, and no Korotkoff sounds are heard.

**Segment B**—Gradual cuff deflation from the reference systolic pressure to the cuff pressure equal to mean pressure (Figure 2).

Turbulent blood flow starts passing under the cuff into the vasculature of the hand. Phase I Korotkoff sounds are heard. The relationship between blood flow ( $Q$ ), resistance of the vasculature ( $R$ ), and pressure ( $P$ ) is expressed<sup>12</sup> as

$$Q = P / R.$$

The hand vasculature initially exhibits low resistance to the flow. The energy escaping into the hand counteracts vascular unloading and the slope of oscillometric waveform envelope decreases. As more flow passes past the cuff, blood volume and pressure in the hand increase due to blocked venous return. The PPG tracing reflects the rising volume and pressure in the hand by rising baseline and increasing waveform amplitudes. When cuff pressure and arterial wall pressure are equal, the oscillometric waveforms reach the maximum amplitude. The cuff pressure at this point is equal to mean pressure. The oscillometric waveform contours are distorted because of

the continuing partial occlusion of the artery. Blood flow is still turbulent and Korotkoff sounds are present.

**Segment C**—Gradual cuff deflation from mean pressure to the reference diastolic pressure (Figure 2).

The oscillometric waveform amplitudes start decreasing with decreasing cuff pressures according to vascular unloading. Blood flow is still turbulent. Korotkoff sounds become muffled when cuff pressure approaches the reference diastolic pressure value (Phase IV). The oscillometric waveform contours continue to be distorted because the artery is still partially occluded. The PPG waveform contours are also distorted. When cuff pressure is at the reference diastolic pressure, the flow becomes laminar and Korotkoff sounds are no longer heard (Phase V). The artery under the cuff is free from occlusion and the oscillometric and PPG waveform contours are no longer distorted.

**Segment D**—Gradual cuff deflation from the reference diastolic pressure to the end of the procedure (Figure 2).

When cuff pressure is further lowered below the reference diastolic pressure, the radial artery under the cuff is free from occlusion, blood flow is laminar, and oscillometric waveforms are not distorted. Korotkoff sounds are not present. Further cuff pressure lowering decreases the oscillometric waveform amplitudes according to vascular unloading. At some arbitrary pressure the cuff is quickly deflated and the deflation procedure is terminated.

Observations of the effects of blood flow under the cuff and in the hand on the oscillometric waveform amplitude envelope resulted in the following hypothesis: The slope of oscillometric waveform amplitude envelope at cuff pressures higher than the reference systolic pressure and the slope at cuff pressures between mean pressure and reference diastolic pressure are steeper than the slope between reference systolic pressure and mean pressure.

To test the hypothesis, three slopes ( $S1$ – $S3$ ) on the waveform amplitude envelope were computed and compared.  $S1$  is the slope from cuff pressure 30 mmHg higher than reference systolic pressure to the cuff pressure equal to the reference systolic pressure.  $S2$  is the slope from cuff pressure equal to the reference systolic pressure to the cuff pressure equal to mean pressure.

$S3$  is the slope from cuff pressure equal to mean pressure to cuff pressure equal to reference diastolic pressure.

$$S1 = (WA_{HSBP} - WA_{SBP}) / (CP_{HSBP} - CP_{SBP}) \quad (1)$$

$$S2 = (WA_{SBP} - WA_{MAP}) / (CP_{SBP} - CP_{MAP}) \quad (2)$$

$$S3 = (WA_{MAP} - WA_{DBP}) / (CP_{MAP} - CP_{DBP}) \quad (3)$$

**Table 1.** Measured and computed data from 32 subjects acquired during gradual cuff deflation procedure.  $SBP_{REF}$  is reference systolic pressure measured by auscultation.  $DBP_{REF}$  is reference diastolic pressure measured by auscultation. MAP is mean arterial pressure determined by a software algorithm. S1, S2, and S3 are oscillometric waveform amplitude envelope slopes computed according to Formulas 1–3.

| Subject            | $SBP_{REF}$ | MAP | $DBP_{REF}$ | S1     | S2     | S3    |
|--------------------|-------------|-----|-------------|--------|--------|-------|
| 1                  | 128         | 96  | 81          | -0.070 | -0.048 | 0.142 |
| 2                  | 146         | 103 | 80          | -0.052 | -0.021 | 0.130 |
| 3                  | 141         | 111 | 96          | -0.060 | -0.031 | 0.160 |
| 4                  | 113         | 90  | 72          | -0.090 | -0.022 | 0.135 |
| 5                  | 130         | 103 | 88          | -0.069 | -0.029 | 0.090 |
| 6                  | 136         | 107 | 95          | -0.079 | -0.030 | 0.099 |
| 7                  | 146         | 105 | 83          | -0.041 | -0.016 | 0.068 |
| 8                  | 158         | 120 | 86          | -0.099 | -0.041 | 0.073 |
| 9                  | 154         | 122 | 102         | -0.045 | -0.023 | 0.080 |
| 10                 | 103         | 83  | 68          | -0.073 | -0.035 | 0.080 |
| 11                 | 147         | 112 | 95          | -0.077 | -0.006 | 0.109 |
| 12                 | 143         | 102 | 83          | -0.084 | -0.028 | 0.153 |
| 13                 | 159         | 119 | 96          | -0.064 | -0.060 | 0.123 |
| 14                 | 125         | 100 | 82          | -0.052 | -0.027 | 0.126 |
| 15                 | 120         | 93  | 78          | -0.062 | -0.017 | 0.151 |
| 16                 | 129         | 103 | 89          | -0.077 | -0.025 | 0.096 |
| 17                 | 125         | 98  | 83          | -0.099 | -0.035 | 0.121 |
| 18                 | 142         | 116 | 100         | -0.027 | -0.005 | 0.069 |
| 19                 | 152         | 124 | 106         | -0.041 | -0.013 | 0.069 |
| 20                 | 155         | 112 | 95          | -0.113 | -0.034 | 0.209 |
| 21                 | 107         | 89  | 72          | -0.097 | -0.015 | 0.116 |
| 22                 | 100         | 74  | 60          | -0.083 | -0.038 | 0.119 |
| 23                 | 106         | 79  | 63          | -0.059 | -0.017 | 0.148 |
| 24                 | 145         | 113 | 93          | -0.097 | -0.037 | 0.151 |
| 25                 | 116         | 87  | 73          | -0.081 | -0.015 | 0.140 |
| 26                 | 145         | 113 | 93          | -0.023 | -0.016 | 0.066 |
| 27                 | 135         | 97  | 82          | -0.069 | -0.018 | 0.141 |
| 28                 | 104         | 80  | 65          | -0.045 | -0.007 | 0.133 |
| 29                 | 146         | 117 | 100         | -0.049 | -0.028 | 0.100 |
| 30                 | 112         | 90  | 74          | -0.032 | -0.025 | 0.070 |
| 31                 | 144         | 104 | 85          | -0.041 | -0.013 | 0.107 |
| 32                 | 134         | 103 | 88          | -0.045 | -0.012 | 0.068 |
| Mean               | 132         | 102 | 85          | -0.065 | -0.025 | 0.114 |
| Standard Deviation | 17          | 13  | 12          | 0.022  | 0.012  | 0.035 |

$WA_{HSBP}$  is the wave amplitude at cuff pressure about 30 mmHg higher ( $CP_{HSBP}$ ) than cuff pressure at reference systolic pressure.  $WA_{SBP}$  is the wave amplitude at cuff pressure equal to the reference systolic pressure ( $CP_{SBP}$ ).  $WA_{MAP}$  is the wave amplitude at cuff pressure equal to the computed mean pressure ( $CP_{MAP}$ ).  $WA_{DBP}$  is the wave amplitude at cuff pressure equal to the reference diastolic pressure ( $CP_{DBP}$ ).

The tabulated data are shown in Table 1. The slopes S1, S2, and S3 were computed with Formulas 1–3.

## Discussion

The above observations and experimental data in Table 1 support the central hypothesis. All the S1 and S3 slopes in Table 1 are steeper than the S2 slopes. The intersubject variability suggests that the slopes are affected by a number of variables. Arterial compliance, mean pressure, heart rate, stroke volume, and blood viscosity have been cited as factors affecting the slopes.<sup>7–9</sup> These factors do not change substantially during a single gradual cuff deflation. Our study suggests that the blood flow under the

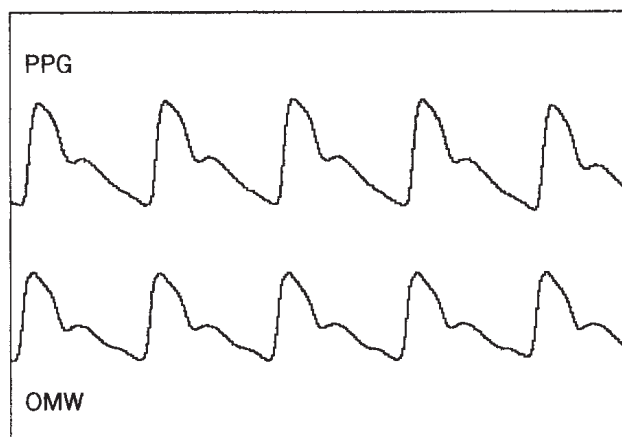
**Table 2.** Measured and computed data from a hand cooling experiment.  $SBP_{REF}$  is reference systolic pressure measured by auscultation.  $DBP_{REF}$  is reference diastolic pressure measured by auscultation. MAP is mean arterial pressure determined by a software algorithm. S2 is oscillometric waveform amplitude envelope slope computed according to Formula 2.

| Hand | $SBP_{REF}$ | MAP | $DBP_{REF}$ | S2     |
|------|-------------|-----|-------------|--------|
| Warm | 140         | 100 | 84          | -0.023 |
| Cold | 139         | 105 | 88          | -0.059 |

cuff and in the hand is an important physiological variable decreasing S2 during a gradual cuff deflation procedure. A simple hand-cooling experiment was conducted to demonstrate that altered blood flow affects S2. The “warm” data were obtained at room temperature of 24°C. The “cold” data were obtained from the same subject after a 3-minute hand cooling with a gel ice pack. Hand cooling constricts smaller blood vessels, increases peripheral resistance, and decreases blood flow. Decreased blood flow results in increased slope S2. The data in Table 2 show the increased S2 after cooling. Changes in the slopes affect the characteristic ratio method of oscillometric systolic pressure determination. A steeper S2 shifts the systolic pressure toward lower values.

This study supports conclusions by theoretical studies that the characteristic ratio method may only be capable of approximate estimation of oscillometric blood pressures.<sup>7-9</sup> Studies of oscillometric monitors under varied clinical conditions revealed deviations that may have been caused by changing slopes.<sup>13-15</sup>

Validation studies usually report larger deviations of systolic values than diastolic values and generally larger deviations of blood pressure values at higher pressures. The observations and the data in this study support the results of validation studies. To illustrate the deviations of blood pressure values, the Geddes ratio algorithm (50% for systolic and 80% for diastolic pressures) was applied to the data from the hand cooling experiment. The resulting warm hand systolic pressure (152 mmHg) is higher than the systolic pressure after hand cooling (145 mmHg). The warm hand diastolic pressure value (87 mmHg) is lower than the diastolic pressure after hand cooling (91 mmHg). An algorithmic method capable of adjusting the ratios according to changing slopes could improve the accuracy of oscillometric systolic and diastolic pressure determination. A method capable of detecting the transition from slope S1 to S2 could improve the accuracy of



**Figure 3.** Finger photoplethysmographic (PPG) waveforms (upper trace) and oscillometric waveforms (OMWs; bottom trace) acquired at the reference diastolic pressure value.

systolic pressure determination. Such methods should be less sensitive to changes of the slopes.

A high level of accuracy may be difficult to achieve with pure oscillometric methods. The slopes are not very steep and they may be difficult to determine without reference blood pressure values. Furthermore, oscillometric waveform amplitudes and contours are affected by a number of factors, such as movement artifacts, deep breathing, tremors, and arrhythmias. Arrhythmias introduce especially difficult problems because their nature and frequency of occurrence are not always apparent.

An improvement in systolic pressure determination may be possible with an external sensor, such as the sensor used in this study. A method using automatic detection of finger PPG waveforms improved accuracy of systolic pressure measurement.<sup>15</sup> The point where oscillometric waveform contour distortions disappear is also the approximate point where the reference diastolic pressure was determined. Automatic detection of the transition from distorted to undistorted waveforms could lead to an improvement in diastolic pressure determination. Such a method would be in agreement with the disappearance of Korotkoff sounds (Phase V). The similarity of oscillometric and PPG waveform contours obtained at reference diastolic pressure is demonstrated in Figure 3. The waveforms are similar because the wrist cuff and the associated pressure circuit essentially form a plethysmograph.<sup>17</sup> Distortion-free waveforms are suitable for studies of methods that reach beyond blood pressure determination. Finger PPG<sup>18</sup> and other finger waveform methods<sup>19</sup> have been studied. Radial artery tonometry has been used to study



arterial pulse waveforms.<sup>20–21</sup> Oscillometric waveforms acquired during automatic blood pressure tests have been used to compute hemodynamic variables.<sup>22–23</sup>

## Study Limitations

The reference blood pressure values were measured sequentially by auscultation from the brachial artery while the oscillometric waveforms were acquired from the wrist. Measurements at different cuff sites have a potential for errors. Because blood flow under the cuff and in the hand was not measured in this study, the alterations of the slopes should be viewed as qualitative rather than quantitative.

## Conclusion

This study suggests that blood flow under the wrist cuff and in the hand affects oscillometric waveforms during oscillometric blood pressure measurement. Algorithmic methods capable of recognizing the changes in envelope slopes and the changes in waveform contours could improve the accuracy of oscillometric blood pressure determination. Such methods would be in agreement with Korotkoff sounds and with plethysmography. The observations of oscillometric and photoplethysmographic waveform contours point to applications reaching beyond blood pressure measurement. ■

## References

1. Marey EJ. *Pression et Vitesse du Sang*. Masson, Paris, France: Physiologie Experimentale; 1876.
2. Posey JA, Geddes LA. The meaning of the point of maximum oscillations in the cuff pressure in the indirect measurement of blood pressure. *Cardiovasc Res Bull*. 1969;8:15–25.
3. Geddes LA, Voelz M, Combs C, Reiner D, Babbs CF. Characterization of the oscillometric method for measuring indirect blood pressure. *Ann Biomed Eng*. 1982;10:271–280.
4. Borow KM, Newburger JW. Noninvasive estimation of central aortic pressure using the oscillometric method for analyzing systemic artery pulsatile blood flow: Comparative study of indirect systolic, diastolic and mean brachial artery pressure with simultaneous direct ascending aortic pressure measurements. *Am Heart J*. 1982;103:879.
5. Ng KG. Blood pressure measurement. *Med Electronics*. February 1999;8:61–64.
6. Staessen JA. Blood pressure-measuring devices: Time to open Pandora's box and regulate. *Hypertension*. 2000;35:1037.
7. Forster FK, Turney D. Oscillometric determination of diastolic, mean and systolic blood pressure. A numerical model. *ASME J Biomech Eng*. 1986;108:359–364.
8. Drzewiecki G, Bansal V, Keram E, Hood R, Apple H. Mechanics of the occlusive arm cuff and its application as a volume sensor. *IEEE Trans Biomed Eng*. 1993;40:704–708.
9. Ursino M, Cristalli C. A mathematical study of some biomechanical factors affecting the oscillometric blood pressure measurement. *IEEE Trans Biomed Eng*. 1996;43:761–776.
10. Jones DW, Appel LJ, Sheps SG, Roccella EJ, Lenfant C. Measuring blood pressure accurately. *JAMA*. 2003;289:1027–1030.
11. Jilek J, Fukushima T. Oscillometric blood pressure measurement: The methodology, some observations, and suggestions. *Biomed Instrum Technol*. 2005;39:237–241.
12. Berne RM, Levy MN. *Cardiovascular Physiology*, Mosby-Year Book. St. Louis, Mo: Mosby; 1992.
13. Czarkowski M, Mikulska M, Zebrowski M, Baran A, Rozanowski K. Venous blood redistribution alters the accuracy of Omron HEM-705CP. *Blood Press Monit*. 2004;9:9–12.
14. vanPopele NM, deBeer N, van der Kuip D, Hofman A, Grobbee DE, Witteman J. Arterial stiffness as underlying mechanism of disagreement between an oscillometric blood pressure monitor and a sphygmomanometer. *Hypertension*. 2000;36:484–488.
15. Quinn M. Automated blood pressure measurement devices: A potential source of morbidity in preeclampsia? *Am J Obstet Gynecol*. 1994;170:1303–1307.
16. Jilek J, Stork M. Improved noninvasive systolic blood pressure detection with finger photoplethysmograph. *Proceedings of Applied Electronics International Conference*. September 8–9, 2004; Pilsen, Czech Republic: University of West Bohemia 2004.
17. Fronek A. *Noninvasive Diagnostics in Vascular Disease*. New York, NY: McGraw-Hill; 1989.
18. Takazawa K, Tanaka N, Fujita M, Matsuoka O, Saiki T, Aikawa M, et al. Assessment of vasoactive agents by the second derivative of photoplethysmogram waveform. *Hypertension*. 1998;32:365–370.
19. Hon EH, Fukushima T, Park JM, Jilek J. Apparatus for assessing the maternal cardiovascular system during pregnancy. *J Mat Fetal Med*. 1993;2:97–102.
20. Smulyan H, Siddiqui DS, Carlson RJ, London GM, Safar ME. Clinical utility of aortic pulses and pressures calculated from applanated radial-artery pulses. *Hypertension*. 2003;42:150–157.
21. Cameron JD, McGrath BP, Dart AM. Use of radial artery applanation tonometry and a generalized transfer function to determine aortic pressure augmentation in subjects with treated hypertension. *J Am Coll Cardiol*. 1998;32:1214–1220.
22. Germano G, Angotti S, Muscolo M, D'Auria F, Giordano M. The (dP/dt) max derived from arterial pulse waveforms during 24 h blood pressure oscillometric recording. *Blood Press Monit*. 1998;3:213–216.
23. Jilek J, Stork M. An experimental system for estimation of blood pressures and hemodynamics from oscillometric waveforms. *Proceedings of Applied Electronics 2003 International Conference*. September 10–11, 2003; Pilsen, Czech Republic: University of West Bohemia 2003.

From the Department of Obstetrics and Gynecology, King-Drew Medical Center, Los Angeles, CA. Address correspondence to Mr. Jilek, 12021 South Wilmingon Ave, Room 2005, Los Angeles, CA 90059 (e-mail: jilekj@usa.net). Telephone: 310-668-4625.